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RESEARCH ARTICLE

Numerical Behavior of the Riemann Zeta Function Using Real-to-Complex Conversion

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ARTICLE HISTORY

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ABSTRACT

The Riemann zeta function $\zeta(s)$ and its zeros lie at the core of modern analytic number theory. On the critical line $\Re(s) = \frac{1}{2}$, the Hardy function

$$Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right)$$

is real valued and encodes the nontrivial zeros as sign changes. Classical large-scale computations rely heavily on Gram points, Gram intervals, and bracketing strategies for locating zeros.

In this paper we develop a different approach for locating zeros of $Z(t)$ based on a real-to-complex conversion

$$t = \sqrt{N - \frac{1}{4}},$$

which parametrizes large heights t by a real variable N . We combine this parametrization with the Riemann–Siegel formula and Gabcke-type remainder estimates, and introduce a “valley scanner” that detects local minima of $|Z(t)|$ along the resulting trajectory in t -space. Candidate minima are then refined by safeguarded Newton iterations.

We report computations reaching heights up to $t \approx 10^{20}$ and compare our results with existing zero tables, observing excellent agreement in regions with published data and consistent behavior in previously unexplored windows. The method provides an alternative to Gram-point based bracketing that emphasizes the local geometric structure of $Z(t)$ rather than traditional sign-change accounting, and offers a flexible computational tool for further large-scale experiments on the critical line.

KEYWORDS

Riemann zeta function; Hardy Z -function; Riemann–Siegel formula; high-precision computation; numerical analysis; experimental mathematics; real-to-complex mapping; zero-finding algorithms

Primary 11M06; Secondary 11M26, 65B10, 65Y20

1. Introduction

The numerical exploration of the Riemann zeta function $\zeta(s)$ on the critical line $s = \frac{1}{2} + it$ remains a central component of analytic number theory. Large-scale com-

putations of its nontrivial zeros have historically relied on Gram points, Gram intervals, and associated bracketing techniques developed by Titchmarsh, van de Lune and collaborators, and, more recently, Odlyzko. These methods have proven effective for locating and verifying zeros at extremely large heights, but they also introduce structural constraints: the distribution of Gram points becomes increasingly irregular, Gram intervals may fail to bracket a zero, and the analysis often requires delicate case-by-case logic.

This paper introduces an alternative numerical strategy based on a real-to-complex transformation that maps a real parameter N to a corresponding height t on the critical line. Although the transformation is elementary, it induces a geometric structure that interacts favorably with the oscillatory behavior of the Hardy function $Z(t)$. In particular, plotting $|Z(t)|$ under this parametrization reveals a landscape of mountains and valleys whose local minima naturally correspond to neighborhoods of zeta zeros. This observation motivates the construction of a *valley scanner*: a procedure that detects candidate minima of $|Z(t)|$ and uses these points as starting positions for Newton refinements.

The computational framework developed here combines:

- the real-to-complex parametrization $N \mapsto t$,
- high-precision evaluation of the Riemann–Siegel formula,
- Gabcke-type bounds for the remainder term,
- a safeguarded Newton refinement.

Extensive computations were performed for heights up to $t \approx 10^{20}$, with results compared against existing zero datasets. The method consistently reproduces known zeros and displays stable behavior well beyond the range of public tables. Because it does not rely on Gram intervals, the approach offers a geometry-driven alternative to classical bracketing and may provide a complementary viewpoint for future large-scale studies of the zeta function.

The algorithms, datasets, and validation tools used in this study have been publicly released for reproducibility through a persistent DOI. All numerical results reported here were generated using these open resources.

2. Mathematical Background

2.1. The Riemann Zeta Function

The Riemann zeta function is defined for complex arguments $s = \sigma + it$ with $\sigma > 1$ by the absolutely convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Through analytic continuation, this definition extends to all $s \neq 1$. A central identity governing its structure is the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s),$$

which reflects values across the critical line. The nontrivial zeros of $\zeta(s)$ lie in the strip $0 < \sigma < 1$. The Riemann Hypothesis asserts that all such zeros satisfy

$$\Re(s) = \frac{1}{2}.$$

2.2. Relation to Prime Numbers

The zeta function encodes the multiplicative structure of the positive integers through Euler's product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \quad \Re(s) > 1.$$

Its zeros influence the distribution of primes via the explicit formula relating $\pi(x)$ to the nontrivial zeros $\rho = \frac{1}{2} + it_\rho$:

$$\pi(x) \sim \text{Li}(x) - \sum_{\rho} \text{Li}(x^\rho) + (\text{lower-order terms}).$$

This deep connection motivates the search for efficient and accurate methods for computing zeros at large heights.

2.3. The Hardy Function $Z(t)$

On the critical line, the real-valued Hardy function is defined by

$$Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right),$$

where the Riemann–Siegel theta function is

$$\theta(t) = \arg\left(\Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right) - \frac{t}{2} \log \pi.$$

The multiplicative factor $e^{i\theta(t)}$ ensures that $Z(t)$ is real for real t . The magnitude $|Z(t)|$ exhibits a distinctive mountain–valley structure, and its zeros correspond precisely to the nontrivial zeros of $\zeta(s)$ on the critical line.

2.4. Average Zero Spacing and Density

The Riemann–von Mangoldt formula states that the number of nontrivial zeros with imaginary part in $(0, T)$ is

$$N(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T).$$

Thus, the average spacing between consecutive zeros near height T is

$$\Delta T_{\text{avg}} \approx \frac{2\pi}{\log(T/2\pi)}.$$

This estimate provides a meaningful benchmark for assessing the density and distribution of numerically computed zeros.

2.5. Known Computational Approaches

The numerical evaluation of $Z(t)$ and the computation of its zeros have led to a wide range of techniques, from asymptotic expansions to FFT-based algorithms.

The Riemann–Siegel Formula

For large t , the Dirichlet series representation of $\zeta(s)$ converges too slowly to be practical. The Riemann–Siegel formula provides the asymptotic expansion

$$Z(t) = 2 \sum_{n=1}^N \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} + R(t), \quad N = \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor,$$

with a small remainder term $R(t)$. Gabcke’s refinements improve the evaluation of the remainder and help ensure numerical stability at large heights.

Gram Points

Gram points are values g_n defined by

$$\theta(g_n) = n\pi.$$

Empirically, zeros of $Z(t)$ tend to alternate with Gram points, although Gram’s law fails infinitely often. Nonetheless, Gram points remain a classical tool for locating sign changes.

Turing’s Method

Turing introduced a verification procedure [4] that ensures no zeros are missed in an interval by comparing the number of sign changes with the theoretical count predicted by the Riemann–von Mangoldt formula. This remains the standard completeness check in computational work.

Odlyzko–Schönhage Algorithm

At very large heights, the Odlyzko–Schönhage [5] method uses FFT-based acceleration to evaluate $Z(t)$ at many points simultaneously. This approach enabled computations of millions of zeros far beyond 10^{20} and remains the state of the art for ultra-high-range verification.

Limitations at High t

As t increases:

- the oscillation of $Z(t)$ intensifies, increasing the required working precision;
- the Riemann–Siegel expansion becomes increasingly sensitive to remainder terms;

- FFT-based methods face memory-scaling limitations and difficulties integrating arbitrary precision.

These constraints motivate alternative formulations aimed at numerical stability and conceptual simplicity, such as the real-to-complex conversion and valley-scanning methods developed in this work.

3. Proposed Numerical Method

3.1. Motivation and Research Shift

Although the present work focuses on the Riemann zeta function and the numerical detection of its zeros, the underlying method originated from an independent study of semiprime numbers. In that earlier investigation, the goal was to analyze algebraic relationships between a semiprime $N = p_1 p_2$ and its constituent prime factors by examining symmetry structures embedded in associated quadratic forms.

Given

$$N = p_1 p_2, \quad S = p_1 + p_2,$$

the classical quadratic equation

$$x^2 - Sx + N = 0$$

has roots p_1 and p_2 . In standard notation, the solutions are

$$x = \frac{S \pm \sqrt{S^2 - 4N}}{2}.$$

While this framework is algebraically natural, it proved difficult to extend into an efficient computational method for factor analysis and was eventually set aside. Nevertheless, the structure revealed by the quadratic form led to a more fruitful perspective.

3.2. N-to-Complex Conversion

Instead of focusing on the two unknown primes p_1 and p_2 , the analysis shifted toward their arithmetic mean

$$m = \frac{p_1 + p_2}{2}.$$

Since $p_2 = 2m - p_1$, substitution into $N = p_1 p_2$ yields

$$N = p_1(2m - p_1),$$

which rearranges into the quadratic equation

$$p_1^2 - 2mp_1 + N = 0.$$

Applying the quadratic formula gives

$$p_1 = m \pm \sqrt{m^2 - N}.$$

While originally intended to study real-valued behavior such as the area under the quadratic curve, numerical experiments showed no stable correlation between m , the discriminant, and the underlying semiprime N . This led to an alternative interpretation: whenever $m^2 < N$, the expression naturally produces complex-valued solutions. Reformulated, the roots become

$$p_1 = m \pm i\sqrt{N - m^2},$$

which suggests treating the quadratic structure as a mapping that associates each real input N with a complex value.

This led to the question: *What happens if the mean is fixed at a known value?* In particular, setting

$$m = \frac{1}{2}$$

aligns the construction with the critical line of the Riemann zeta function. With this choice, the expression becomes

$$p(N) = \frac{1}{2} \pm i\sqrt{N - \frac{1}{4}},$$

placing the real part exactly at $\Re(s) = \frac{1}{2}$.

Thus the semiprime-motivated quadratic identity becomes a simple but powerful real-to-complex conversion:

$$N \mapsto s(N) = \frac{1}{2} \pm i\sqrt{N - \frac{1}{4}}.$$

Each real input N is mapped to a point on the critical line, with imaginary part determined by the square-root term. Numerical exploration of this mapping suggested structured behavior in the corresponding evaluations of the Hardy Z -function, ultimately guiding the development of the valley-scanner algorithm.

3.3. Valley Scanner Algorithm

The term *valley scanner* refers to the numerical procedure used to identify zeros of the Hardy function by detecting local minima of $|Z(t)|$ along a parameterized traversal of the critical line. The construction begins with the mapping

$$s = m \pm i\sqrt{N - m^2}, \quad m = \frac{1}{2},$$

so that

$$t = \Im(s) = \sqrt{N - m^2}, \quad N = t^2 + m^2.$$

Thus increments in N generate corresponding increments in the imaginary part t , providing a simple mechanism for scanning $|Z(t)|$ across a continuous range.

The objective is to evaluate $|Z(t)|$ for successive values of N and analyze the numerical landscape through direct observation and visualization.

A Python script was developed for the initial experimental exploration. The formatted version is available in the public repository: [playground.py](#). The results below reproduce the exploratory code.

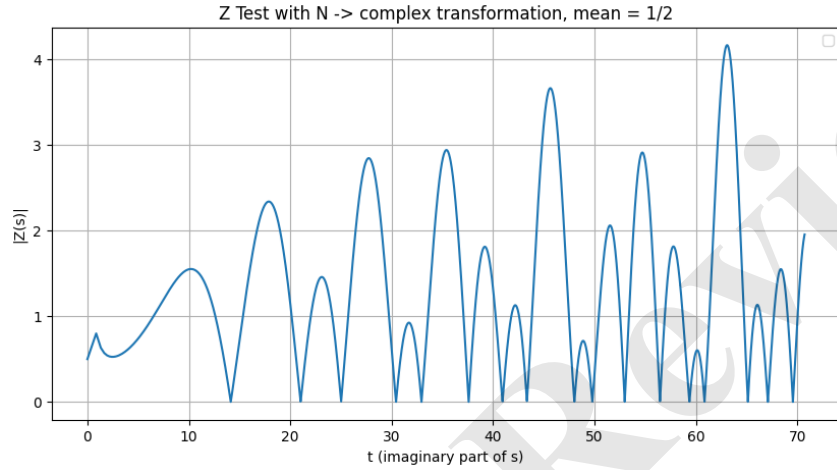


Figure 1. Illustration of the valley scanner detecting local minima of $|Z(s)|$.

A comparison with the first nontrivial zeros of the Riemann zeta function is shown in Table 1.

Table 1. First nontrivial zeros of the Riemann zeta function.

Index	Imaginary Part (t_n)
1	14.1347251417346937904572519835625
2	21.0220396387715549926284795938969
3	25.0108575801456887632137909925628
4	30.4248761258595132103118975305840
5	32.9350615877391896906623689640747
6	37.5861781588256712572177634807053
7	40.9187190121474951873981269146334
8	43.3270732809149995194961221654068
9	48.0051508811671597279424727494277
10	49.7738324776723021819167846785638
11	52.9703214777144606441472966088808
12	56.4462476970633948043677594767060
13	59.3470440026023530796536486749922
14	60.8317785246098098442599018245241
15	65.1125440480816066608750542531836
16	67.0798105294941737144788288965221
17	69.5464017111739792529268575265547

Visually, the resulting plot is self-explanatory. When the Riemann zeta function $\zeta(s)$ is evaluated using the parametrization $s = m \pm i\sqrt{N - m^2}$ with $m = \frac{1}{2}$, and with N ranging over an arbitrary numerical interval, the computed values consistently approach zero whenever the corresponding height t is near a true zero of the zeta function. This behavior is fully consistent with the theoretical expectation that the transformation traces the critical line and that the magnitude of $\zeta(s)$ should decrease in the vicinity of its nontrivial zeros.

Refining the range of N by introducing decimal-level resolutions produces progressively closer approximations to the actual zeros. The accompanying Python script reproduces this behavior across any chosen interval of N , though the computations become increasingly demanding as the height t grows.

The dataset obtained from this exploratory script is summarized below and compared against confirmed zeros of the Riemann zeta function.

Table 2. Comparison between computed and reference zeros of the Riemann zeta function.

N	Computed t	Reference t
200	14.13329402510257	14.13472514
442	21.017849556983705	21.02203964
626	25.014995502697975	25.01085758
926	30.426140077242792	30.42487613
1085	32.93554311074891	32.93506159
1413	37.586566749305526	37.58617816
1675	40.92370951	40.91871901
1877	43.32147273581544	43.32707328
2305	48.00781186432057	48.00515088
2478	49.77700272214067	49.77383248
2806	52.96933074902875	52.97032148
3186	56.442448564887755	56.44624770
3522	59.34433418617145	59.34704400
3701	60.83378995	60.83177853
4240	65.11336268386084	65.11254405
4500	67.08017591	67.07981053
4837	69.54674686856315	69.54640171

3.4. Identifying Zeros via Consecutive Minima of $|Z(t)|$

Although Python provides a flexible environment for preliminary experiments, the valley-scanning approach becomes computationally intensive as the height t increases. To explore larger regions of the critical line, optimized implementations were developed in C++ with arbitrary-precision arithmetic, together with parallel execution on AWS EC2 instances.

The underlying idea remains unchanged: the magnitude $|Z(t)|$ exhibits an alternating pattern of peaks and valleys, and the nontrivial zeros occur precisely at the bottoms of these valleys. The algorithm therefore scans successive values of N (and the corresponding t) to detect transitions from descent to ascent in the numerical landscape.

Mountain walking and valley detection

The valley-scanner algorithm follows a simple geometric principle:

- Choose an initial height t_0 .
- Convert t_0 to its corresponding value N_0 via $N = t^2 + m^2$.
- Increment N by one unit (or by a chosen step size).
- Compute the resulting t_1 and evaluate $|Z(t_1)|$.
- If $|Z(t_1)| < |Z(t_0)|$, the scan is descending; otherwise it is ascending.
- A change from descent to ascent indicates a local minimum and therefore a candidate zero.
- Each candidate is refined by a Newton root-finding method.

This mechanism serves as the foundation for the large-scale computations described later in the paper.

Evaluation of the Hardy function

The numerical evaluation of $Z(t)$ relies on several components:

- The Riemann–Siegel formula

$$Z(t) = 2 \sum_{n=1}^N \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} + R(t),$$

where $N = \lfloor \sqrt{t/(2\pi)} \rfloor$, and $R(t)$ is the remainder term estimated using Gabcke-type refinements.

- The phase function $\theta(t)$ is computed via a high-precision evaluation of $\log \Gamma(\frac{1}{4} + \frac{it}{2})$ using Stirling expansions:

$$\theta(t) = \text{Im} \left[\log \Gamma \left(\frac{1}{4} + \frac{it}{2} \right) \right] - \frac{t}{2} \log \pi.$$

- Zero candidates are refined using a safeguarded Newton method with a Pegasus-type fallback (a hybrid Newton–secant scheme). Computations are performed at 128–256 bits of precision to stabilize convergence in highly oscillatory regions.
- Validation is performed by comparing the refined zeros with independent datasets and by applying a local consistency test (the “ball test”) across neighboring zeros.

3.5. Parallelization and Cloud Execution

The dominant computational cost in evaluating $Z(t)$ through the Riemann–Siegel formula arises from the main summation

$$Z(t) = 2 \sum_{n=1}^N \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} + R(t).$$

For fixed t , each term $\cos(\theta(t) - t \log n)/\sqrt{n}$ is independent, making the summation highly amenable to parallelization.

We partition the summation interval into subranges

$$S_i = \sum_{n=n_i}^{n_{i+1}-1} \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}},$$

and aggregate the partial sums. The remainder term $R(t)$ is evaluated separately using Gabcke's expansion.

Only the main summation is parallelized; the correction term remains serial. In practice this yields near-linear scaling with the number of physical CPU cores. AWS EC2 instances are configured with OpenMP to distribute work across all cores using deterministic reduction clauses, ensuring numerical stability and reproducibility.

This parallel strategy remains effective even when $N = \sqrt{t/(2\pi)}$ reaches into the millions, as occurs for very large heights.

3.6. Why Spacing Heuristics (and Naive Prefiltering) Can Skip Zeros

The Riemann–von Mangoldt formula

$$N(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T) \quad (1)$$

implies an approximate zero density

$$N'(T) \approx \frac{1}{2\pi} \log\left(\frac{T}{2\pi}\right),$$

and an average spacing

$$s_{\text{avg}}(T) \approx \frac{2\pi}{\log(T/(2\pi))}. \quad (2)$$

3.6.0.1. Key caveat.. While (2) provides a reliable mean estimate, the *local* spacing between consecutive zeros can differ substantially. Advancing t by a deterministic step of size $s_{\text{avg}}(T)$, or rejecting points based on naive thresholds, may cause genuine zeros to be skipped.

3.6.0.2. Concrete example (skipped zero).. Two consecutive zeros verified independently satisfy

$$t_1 = 30607946001.041439, \quad t_2 = 30607946001.175073,$$

yielding a gap

$$\Delta_{\text{true}} = t_2 - t_1 \approx 0.133634.$$

The spacing estimate at $T \approx t_1$ is

$$s_{\text{avg}}(t_1) = \frac{2\pi}{\log(t_1/(2\pi))} \approx 0.281673,$$

which overshoots the next zero by a wide margin.

3.6.0.3. Implication for the valley scanner.. Early variants of the algorithm explored spacing-based stepping and threshold-based prefilters. Both approaches resulted in missed zeros. The final version therefore avoids prefilters and relies instead on dense sampling with minima detection, followed by safeguarded refinement.

3.6.0.4. Why GPU acceleration is not used.. Although the Riemann–Siegel summation is parallel in principle, the large- t computations in this work require 200–500 bits of precision, which current GPUs do not support natively. Emulating multi-precision arithmetic across hundreds of GPU threads introduces significant overhead, and the branching and precision adjustments inherent to $Z(t)$ evaluations conflict with the SIMD execution model. For these reasons, all high-precision evaluations are performed on multi-core CPU instances using MPFR/MPC with OpenMP.

Software and Computational Stack

The computational workflow is built on:

- **C++** with MPFR/MPC for high-precision evaluation of $Z(t)$.
- **AWS EC2** for large multi-core compute instances.
- **AWS Lambda** for batch orchestration and monitoring.
- **AWS S3** for persistent storage of logs and zero tables.
- **AWS Cognito** for secure authentication.
- **Docker** and AWS ECR for reproducible deployments.

An authenticated web interface allows users to run valley scans and visualize zeros up to heights near 10^{16} . While the full source code is not yet archived, all numerical datasets used in this work are publicly available via Zenodo and GitHub. A public Docker image is provided for reproducibility; see the project documentation: [README.md](#).

4. Results

This section presents representative outputs obtained by the valley-scanner framework across several ranges of t , including large heights where refinement is particularly challenging. The analysis of average zero spacing and local fluctuations was cross-checked by independent numerical scripts and symbolic tools to validate the results.

4.0.0.1. Scope of datasets.. The datasets shown here are intentionally concise and primarily illustrative. The valley scanner is designed to detect zeros within any specified range of t , and the batch-processing architecture supports large-scale computation over much broader intervals, with feasibility limited mainly by practical cloud-computing costs.

Table 3. Zero heights near $t \approx 1.1223 \times 10^9$.

t values
1122334455.05585408179597553373792753809670000283010284579445795581
1122334455.76031733677095186350835049571870227573769745536211492284
1122334455.96549341079558436082594904200898020740502977379833207345
1122334456.55122003455096567518805961663152718628678713782068430266
1122334456.69117656282847361682640086578613693696115124531479812624
1122334456.83867869196755030364329563494267434476218915992393583291
1122334457.15371418683945191486723066707952951489968507756091862973
1122334457.51390511989118717254611539418981231215406357632944032390
1122334457.76838741510620113455239734916962707865127304851980893696
1122334458.27155512243026569374168652041355212655972394951814858601
1122334458.44139136841692632939134894613526508309356307629078128927
1122334458.89501425212585279215913084722248689346999249199510723479
1122334458.95374621822277785558114513758006054444953092856654909690
1122334459.47180990048612714347328799643237077235245383445928401233
1122334459.78629512841632863716858891532650291270784909335852103315
1122334460.10222132816663933236459832453382800744621284674207091481
1122334460.34922785530398779220541441366240746601424248348045312059
1122334460.78924267475219010097101478269117727805242937830818188169
1122334461.03129979223085870682939517578522137411650684496149763174
1122334461.34105411839180789869245630301714029644280234966395722460
1122334461.68097513340590425015663066588174538735332969362229273055
1122334462.05653607553267581154859487598541969417453119764160088476
1122334462.35999091678192331701334666903361418145334941591973792237
1122334462.71045060502215878868517734546324575820479730409902219700
1122334463.22570175761036362585961260605296584140311386148887315924

Full dataset:

[refined_sample.1122334455.csv](#)

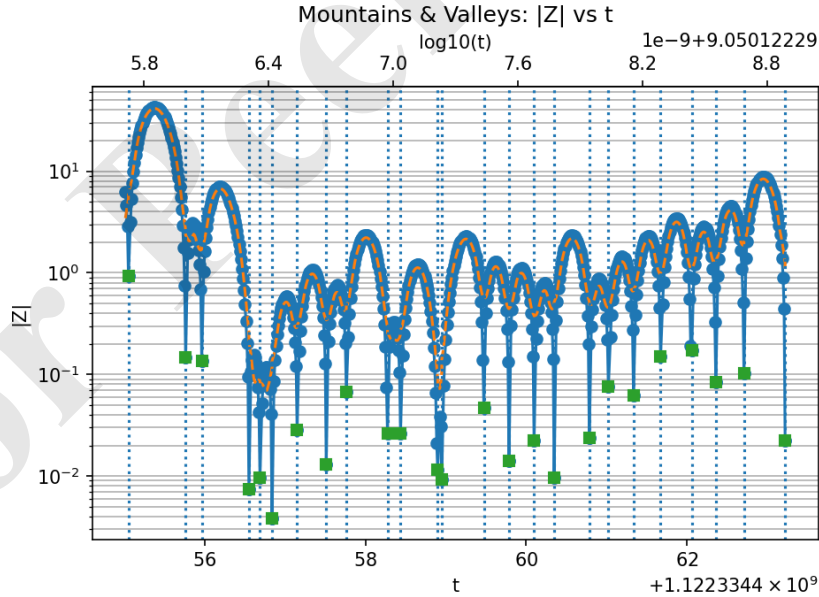


Figure 2. Valley-scanner output near $t \approx 1.1 \times 10^9$; local minima of $|Z(t)|$ are marked prior to refinement.

Table 4. Observed zero spacing vs. theoretical prediction at $t \approx 1.1223 \times 10^9$.

Parameter	Value / Description
Mean height T	1.122334459×10^9
Theoretical formula	$\Delta_{\text{avg}} \approx \frac{2\pi}{\log(T/2\pi)}$
$\log(T/2\pi)$	19.84
Predicted Δ_{avg}	0.33068
Observed mean(Δt)	0.34041
Relative difference	$\approx 2.94\%$

The observed mean spacing differs from the theoretical prediction by about 2.9%, which is consistent with ordinary local fluctuations at this height. This dataset contains twenty-four valid spacings (after discarding the initial $\Delta t = 0$), providing a stable estimate of the zero density.

4.0.0.2. Additional comments..

- The narrowest local gap ($\Delta t \approx 0.05873$) reflects a brief clustering region.
- The widest local gap ($\Delta t \approx 0.70446$) corresponds to a modest sparsity interval.
- Across this range, the valley scanner maintained $|Z(t)|$ residuals below 10^{-17} at refined zeros.

Table 5. Zero heights near $t \approx 7.7888 \times 10^{13}$.

t values
77887788778877.0202277488171921884175366387426290737402889145896393
77887788778877.2496796799624248241926909299697953447734769996091705
77887788778877.3958394524380600874237302170777958741989520806110630
77887788778877.4642294878030461471735144488028421254791094327197845
77887788778877.6527131741588833597118601699816281392973786842778076
77887788778877.9129846403275919999999999999999999999999999999999999
77887788778878.26126506586418260216544072488109368287269084989008
77887788778878.396760708196222789018547628516718575402529917454215242
77887788778878.5408184172663888210849326677030723102974335132064236
77887788778878.8349976024444658352047434901695167999729638964432923

Full dataset:

refined_sample_77887788778878.csv

Table 6. Observed zero spacing vs. theoretical prediction at $t \approx 7.7888 \times 10^{13}$.

Parameter	Value / Description
Mean height T	7.788779×10^{13}
Theoretical formula	$\Delta_{\text{avg}} \approx \frac{2\pi}{\log(T/2\pi)}$
$\log(T/2\pi)$	30.89
Predicted Δ_{avg}	0.20841
Observed mean(Δt)	0.20164
Relative difference	$\approx 3.25\%$

At $t \approx 7.8 \times 10^{13}$, the observed mean spacing lies within about 3.3% of the predicted value, in line with the gradual contraction of Δt as $\log T$ grows and supporting the numerical reliability of the valley scanner at this range.

4.0.0.3. Additional comments..

- The narrowest local gap ($\Delta t \approx 0.06839$) again indicates a short clustering region.
- The widest local gap ($\Delta t \approx 0.34828$) reflects a mild local sparsity.
- The dataset contains nine valid spacings after excluding the initial spacing.

Table 7. Zero heights near $t \approx 2.1212 \times 10^{15}$.

t values
2121212121212121.05717439635565362751782963300490252462746596256208
2121212121212121.42395229690379485836605340059707795745376215022485
2121212121212121.63791118938682163457838808098568955130531660194243
2121212121212121.72223918447411208850522081133640649819206480997688
2121212121212121.93125161293757949737416946721044933465380477203190
2121212121212122.13927614138040154231827268671408632573549285521144
2121212121212122.31254419266807476082116789227597997808943479324778
2121212121212122.58982505389887247687959200860289427367908827950849
2121212121212122.65324071114046090348408431974763989558277572844835
2121212121212122.92082343415959022641562412879095630833398526361303
2121212121212123.14605534867317515793517970162094301498032240314091

Full dataset:

[refined_sample.2121212121212121.csv](#)

Table 8. Observed zero spacing vs. theoretical prediction at $t \approx 2.1212 \times 10^{15}$.

Parameter	Value / Description
Mean height T	2.121212×10^{15}
Theoretical formula	$\Delta_{\text{avg}} \approx \frac{2\pi}{\log(T/2\pi)}$
$\log(T/2\pi)$	34.19
Predicted Δ_{avg}	0.18782
Observed mean(Δt)	0.20889
Relative difference	$\approx 11.2\%$

At $t \approx 2.1 \times 10^{15}$, the observed mean spacing exceeds the theoretical prediction by about 11%. Such over-dispersion is compatible with local density fluctuations observed in large-scale zero computations at extreme heights and is qualitatively in line with random-matrix models for the zeros.

4.0.0.4. Additional comments..

- The narrowest local gap ($\Delta t \approx 0.06342$) indicates normal local clustering.
- The widest local gap ($\Delta t \approx 0.36678$) corresponds to a transient sparsity region.
- The valley scanner resolved ten consecutive zeros at this height without evident numerical drift.

Table 9. Highest evaluated heights from the valley scanner.

Parameter	Value
t	9999999999999999.232850606909407706487068201450796323115355335175991058349609375
$ Z(t) $	$3.657602321013126733848087942217412642846910689346499860361067 \times 10^{-13}$
t	197909211979092119791.004810035047150401470161806827263717423193156719207763671875
$ Z(t) $	$8.258118375241327319604320773664599713029092314507266420324956 \times 10^{-17}$

Remark. The largest height, $t \approx 1.9790921 \times 10^{20}$, was chosen so that its leading digits encode the author's birth date (1979-09-21). This serves as a symbolic benchmark and illustrates that the valley scanner can be directed to specific target heights while maintaining numerical stability at very large t .

5. Discussion

5.1. Summary of observations

Several observations emerge from the numerical experiments:

- The continuous “mountain-valley progression” in $|Z(t)|$ suggests that, within the probed intervals and at the sampling densities used, no zeros are skipped. Occasional minor irregularities (small intermediary peaks) do appear, but increasing the resolution in N sharpens the localization of true minima in such regions.
- The mapping $N \mapsto s(N)$ enforces $\Re(s) = \frac{1}{2}$ by construction, so the valley scanner naturally explores the critical line.
- The method is conceptually simple and appears scalable; it could, in principle, be adapted to other L -functions once an appropriate analogue of the $N \rightarrow$ complex mapping is specified.

The primary goal of this work is not to provide yet another large zero dataset, but rather to formalize the $N \rightarrow$ complex conversion (with $m = \frac{1}{2}$) as a computational lens for studying the Hardy Z -function and its mountain-valley structure.

Another goal of the $N \rightarrow$ complex conversion is to investigate whether alternative “mean” $m \neq \frac{1}{2}$ reveal meaningful structure at large heights. To evaluate $\zeta(s)$ away from the critical line, we used standard analytic continuation methods, such as approximate functional equations for $\zeta(s)$, as implemented in high-precision libraries. Our off-line experiments therefore rely on well-established techniques rather than any modification of the valley scanner itself.

Although the exploratory scans performed here (Figures below) did not detect zeros for $m \in \{1/5, 1/\pi, 1/3, 1/e\}$ within the tested ranges, the numerical framework highlights a general possibility: the $N \rightarrow$ complex parametrization may serve as a systematic tool for searching for hypothetical zeros off the critical line, should they exist.

This does not provide evidence for or against such zeros; rather, it illustrates that the conversion offers a unified computational lens for examining both the critical line and nearby vertical lines, and may support future investigations into the global structure of $\zeta(s)$.

5.2. Exploration of alternative mean values

An important conceptual question is whether different choices of the mean parameter m might reveal structure away from the critical line. To probe this, exploratory computations were performed for several representative values:

$$m = \frac{1}{5}, \quad m = \frac{1}{\pi}, \quad m = \frac{1}{3}, \quad m = \frac{1}{e}.$$

In each case, the corresponding mapping $N \mapsto s(N) = m \pm i\sqrt{N - m^2}$ traces a vertical line $\Re(s) = m$. Within the numerical ranges tested for these alternative choices of m , no zeros were detected.

This does not constitute evidence about the existence or nonexistence of zeros off the critical line; the experiments were limited in both range and height. Rather, the outcome is consistent with the classical expectation that nontrivial zeros occur on $\Re(s) = \frac{1}{2}$, and it motivates further analytical work to understand why the case $m = \frac{1}{2}$ is particularly well aligned with the valley structure of $Z(t)$.

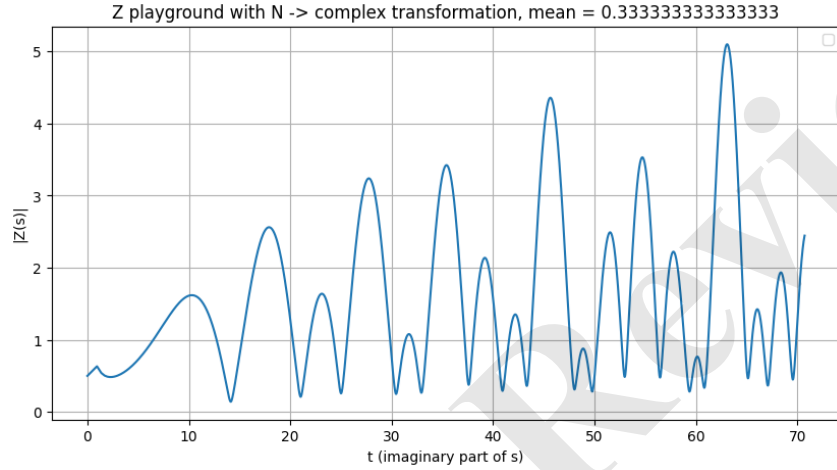


Figure 3. Valley-scan output with $m = 1/3$. No zeros were observed in the tested range.

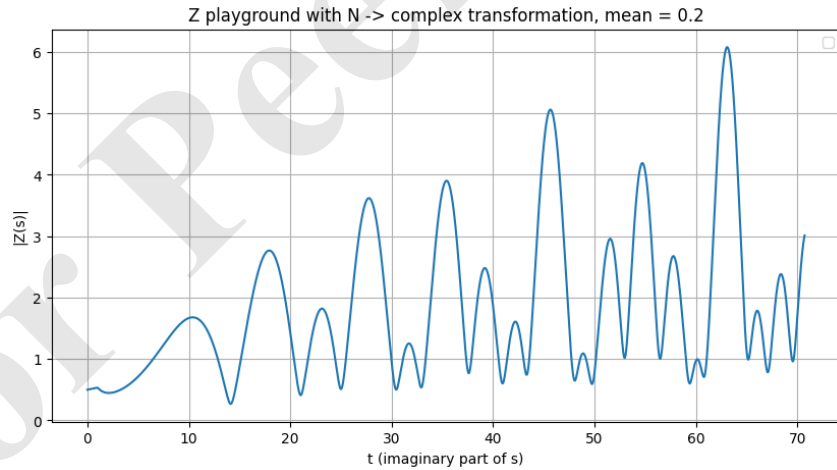


Figure 4. Valley-scan output with $m = 1/5$. No zeros were observed in the tested range.

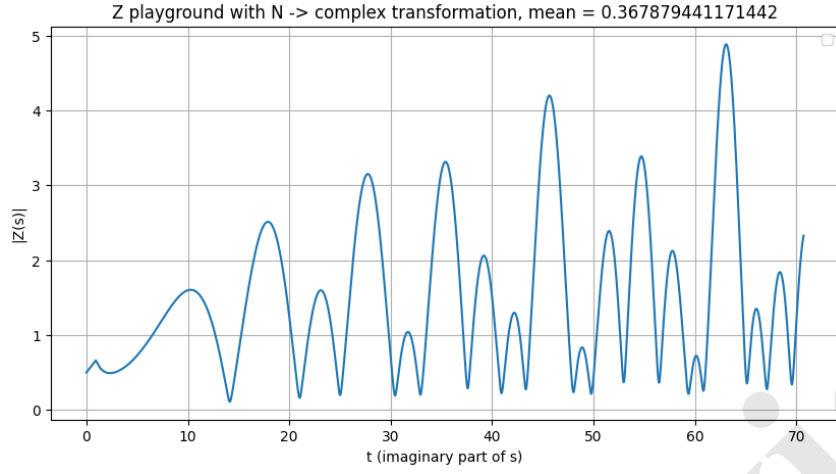


Figure 5. Valley-scan output with $m = 1/e$. No zeros were observed in the tested range.

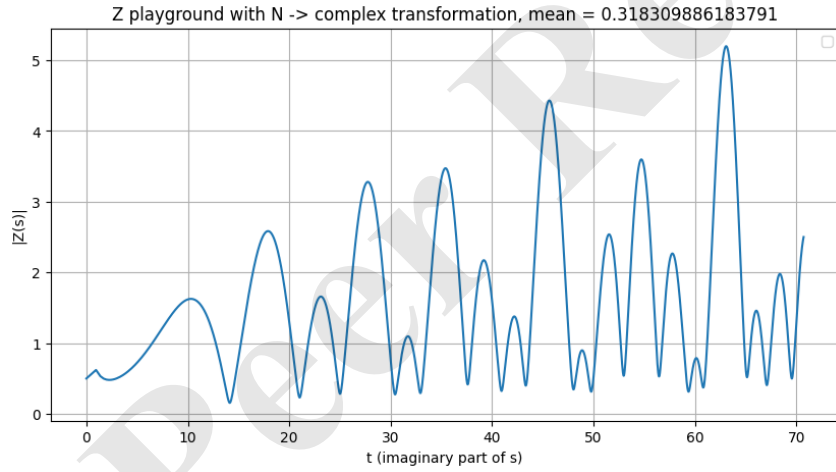


Figure 6. Valley-scan output with $m = 1/\pi$. No zeros were observed in the tested range.

6. Staircase representation of the explicit formula

We give a brief description of a real-valued formulation of the correction term in the explicit formula for the Chebyshev function $\psi(x)$, expressed as partial sums over the nontrivial zeros of the Riemann zeta function. Although this construction is classical, it provides a useful parallel to the “mountain walk” intuition underlying the valley scanner, since both frameworks accumulate oscillatory contributions from the ordinates γ_n .

Let the nontrivial zeros of $\zeta(s)$ be

$$\rho_n = \frac{1}{2} + i\gamma_n, \quad \bar{\rho}_n = \frac{1}{2} - i\gamma_n, \quad \gamma_n > 0.$$

6.1. Staircase partial sums

For fixed $x > 1$, define the complex partial sums

$$L_k(x) = \sum_{n=1}^k \frac{x^{\rho_n}}{\rho_n}, \quad k = 1, 2, \dots, \quad (3)$$

matching the computational form

$$\text{term} = \frac{e^{\rho \log x}}{\rho}, \quad \text{correction} += \text{term}.$$

As k increases, these oscillatory increments build a staircase-like approximation to the full zero contribution in the explicit formula.

6.2. Including conjugate terms

Pairing each zero with its conjugate produces a real-valued staircase:

$$\tilde{L}_k(x) = \sum_{n=1}^k \left(\frac{x^{\rho_n}}{\rho_n} + \frac{x^{\bar{\rho}_n}}{\bar{\rho}_n} \right), \quad (4)$$

represented in code by

$$\text{term} = \frac{e^{\rho \log x}}{\rho}, \quad \text{term2} = \frac{e^{\bar{\rho} \log x}}{\bar{\rho}}, \quad \text{correction} += \text{term} + \text{term2}.$$

6.3. Derivation of the real closed form

Let $L = \log x$. Since

$$x^\rho = x^{1/2} e^{i\gamma L}, \quad \frac{1}{\rho} = \frac{\frac{1}{2} - i\gamma}{\gamma^2 + \frac{1}{4}},$$

a single conjugate pair contributes

$$\begin{aligned} \frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} &= \frac{x^{1/2}}{\gamma^2 + \frac{1}{4}} \left[\left(\frac{1}{2} - i\gamma \right) e^{i\gamma L} + \left(\frac{1}{2} + i\gamma \right) e^{-i\gamma L} \right] \\ &= \frac{x^{1/2}}{\gamma^2 + \frac{1}{4}} \left[\cos(\gamma L) + 2\gamma \sin(\gamma L) \right]. \end{aligned}$$

Thus the real staircase (4) becomes

$$\tilde{L}_k(x) = x^{1/2} \sum_{n=1}^k \frac{\cos(\gamma_n \log x) + 2\gamma_n \sin(\gamma_n \log x)}{\gamma_n^2 + \frac{1}{4}}. \quad (5)$$

6.4. Relation to the standard $-2\Re$ formulation

In the explicit formula it is customary to write the zero contribution as

$$C_k(x) = -2\Re \sum_{n=1}^k \frac{x^{\rho_n}}{\rho_n}.$$

Since $\Re(z) = \frac{1}{2}(z + \bar{z})$, we obtain

$$C_k(x) = -\tilde{L}_k(x),$$

so (5) recovers the standard expression up to the overall sign.

6.5. Rewriting in terms of $N_n = \gamma_n^2 + \frac{1}{4}$

Introduce

$$N_n = \gamma_n^2 + \frac{1}{4},$$

mirroring the “normal numbers” used in the numerical mountain walk. Then (5) becomes

$$\tilde{L}_k(x) = x^{1/2} \sum_{n=1}^k \frac{\cos(\gamma_n \log x) + 2\gamma_n \sin(\gamma_n \log x)}{N_n}. \quad (6)$$

6.6. Computational remarks

Formula (6) removes the need for complex exponentiation: each term requires only real evaluations of $\cos$, $\sin$, and a single logarithm. Thus the staircase can be computed directly from the ordinates γ_n with good numerical stability.

The identity

$$\frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} = \frac{x^{1/2}(\cos(\gamma \log x) + 2\gamma \sin(\gamma \log x))}{\gamma^2 + \frac{1}{4}}$$

serves as a compact bridge between the complex explicit formula and a purely real numerical implementation.

A demonstration script implementing (6) is available at: [staircase.with.real.py](#).

7. Conclusions

The main conclusions of this work may be summarized as follows:

- (1) The real-to-complex mapping

$$N \mapsto s(N) = \frac{1}{2} \pm i\sqrt{N - \frac{1}{4}},$$

combined with high-precision evaluation of the Hardy function via the Riemann–Siegel formula (with Gabcke-type [3] remainder control) and safeguarded Newton refinement, successfully reproduces zeros of $\zeta(s)$ across a broad range of heights—from classical low-lying zeros up to values on the order of 10^{20} .

- (2) The numerical experiments indicate that this formulation offers a conceptually simple computational lens for exploring the zeta function. The symmetry $\Re(s) = \frac{1}{2}$ is built directly into the mapping, and the resulting progression of valleys in $|Z(t)|$ provides a natural mechanism for locating candidate zeros.
- (3) The staircase representation developed in this work expresses the explicit-formula correction term entirely in real operations, depending only on the ordinates γ_n and the quantities $N_n = \gamma_n^2 + \frac{1}{4}$. This mirrors the structure exploited by the valley scanner and may generalize to other L -functions once analogous mappings and conjugate-pair identities are identified.
- (4) While this study does not address the Riemann Hypothesis, it contributes a reproducible numerical framework, high-precision datasets, and a computational methodology that may support further investigations into the structure of $Z(t)$ and the distribution of zeros.

8. Future Work

- (1) The symmetry of the observed “mountain” structures in $|Z(t)|$ suggests that local maxima may encode geometric information about neighboring zeros. Early experiments show that the horizontal distance between a refined zero and its adjacent peak often serves as a reasonable predictor for the next zero. A systematic study of this peak-to-zero geometry could reduce the computational cost of large scans.
- (2) The present work was carried out independently and relies heavily on numerical experimentation. Further development would benefit from collaboration with specialists in analytic number theory and numerical analysis, particularly regarding the theoretical underpinnings of the $N \rightarrow$ complex parametrization and the stability behavior of the valley-scanner across extremely large ranges of t . Additional computational resources would also enable deeper exploration.

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All conceptual ideas, algorithms, numerical experiments, and interpretations presented in this work were developed and validated by the author. AI systems served only as assistants for improving clarity, ensuring consistency, and accelerating routine editorial tasks.

Data and software availability

All source code, Docker images, datasets, and reproducibility materials supporting this study are archived at:

DOI: [10.5281/zenodo.17566257](https://doi.org/10.5281/zenodo.17566257).

The Zenodo record links to the GitHub repository, which includes instructions for running the valley scanner, the web interface, and all auxiliary numerical scripts.

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